

Chimera_OS - LEFT manual

Version: 2026-09-18. Hardware: diceFriend LEFT, Teensy 4.0.

Character and modes

Chimera_OS combines rungler-style registers, FM/PM, pulsars, feedback memory and a low-pass-gate model. AS runs one cybernetic organism and lets separate random dividers select one event every 2-5 source steps. TS runs three independent synthesis instances; pads and MIDI notes in TS do not use the AS 2-5 divider.

Panel and DICE

Hold unshifted DICE for 3 seconds to change AS/TS. Short DICE randomizes all 12 parameters of the active mode, resets event dividers and produces a strike. SHIFT + short DICE resets the state and resumes without randomizing the panel. Hold SHIFT + DICE for 10 seconds to enter the bootloader. A local short DICE also re-arms local Clock Out after incoming MIDI Clock has latched it off.

Knob	AS normal layer	AS SHIFT layer
P1	Root	Coupling: FM/PM and feedback
P2	Modulator ratio	Memory time
P3	Ecology: oscillator → rungler → pulsar	Mutation
P4	Fold	LPG decay time
P5	Aperture: pulse width and LPG brightness	Homeostasis
P6	Density: base clock, about 0.328-590.49 Hz	Space

Knob	TS normal layer	TS SHIFT layer
P1	Voice/pad 1 pitch	Coupling
P2	Voice/pad 2 pitch	Memory
P3	Voice/pad 3 pitch	Mutation
P4	Voice/pad 1 decay	Fold
P5	Voice/pad 2 decay	Aperture
P6	Voice/pad 3 decay	Space

In AS, T1 is Freeze, T2 Invert and T3 Rupture; a new touch also strikes the organism. In TS, each pad controls its own voice and can hold its gate. After explicit Stop or a mode change, release the pad before playing it again.

CV and USB MIDI

CV1 uses rising edges with a 1 ms filter. CV2 has edge detection plus a smoothed continuous path that shifts pitch and sends CC74. AS gives CV1 and CV2 independent 2-5 dividers; TS sends selected triggers directly to successive voices. CV is not a calibrated 1 V/oct input.

MIDI notes and CC use input channel 1. In AS, the 2-5 divider selects Note On events; a skipped note does not change pitch. In TS, notes rotate across three voices and Note Off or velocity 0 releases the assignment. CC20-31 control the active mode's 12 panel values. CC120 stops sound/transport; CC123 releases notes and resets state.

Local MIDI output on channel 1 sends at most 20 notes per second, with Note Off after 35 ms. CV2 sends CC74 at up to 40 messages per second with a 2/127 change threshold. Physical panel changes and DICE send CC20-31. Received messages are not sent as direct USB Thru.

MIDI Clock uses 24 PPQN. AS applies the 2-5 event divider; TS triggers on every complete step. A note stream suppresses extra clock strikes. Clock priority is MIDI Clock → CV1 → CV2 → internal P6. MIDI expires after about 1.8 seconds; CV remains active for at least 1.8 seconds or roughly three measured periods. Start resets, Continue resumes and Stop stops. Receiving Clock latches local Clock Out off until a local short DICE.

LEDs and first patch

LED 0 shows a local MIDI-output event, LED 1 smoothed CV2, and LED 2/3 audio level L/R.

In AS, start with low Density, Ecology toward the oscillator side and modest Coupling. Explore Fold and Ratio, then increase Memory. In TS, set three pitches and three decays; pad response does not wait for the AS divider.